

Keypoint Based Tampered Image Identification



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Abstract Identifying the tampered regions within an image is a challenging field of study. Images are often altered using editing programs or tools such as Picasa, Photoshop, and others, intending to mislead people or conceal information. Therefore, it is essential to verify the validity of photographs to determine whether they have been altered before extracting any valuable information from them. One specific type of tampering, where an object is intentionally duplicated in the image, is known as copy-move forgery. Based on locating AKAZE keypoints in an image, copy-move forgery area detection has been presented in this paper. The AKAZE features are extracted from the image, and brute force matcher technique is used to match features. In order to group geographically confined keypoints that identify cloned regions or forged portions in a picture, the Birch clustering algorithm is applied to the keypoint locations. Images from the MICC dataset series and CoMoFoD datasets are used for the tests, and the outcomes are compared with those of previously published methods.

Keywords AKAZE · BIRCH · Brute force matching · Image tampering

1 Introduction

Today millions of images are captured using digital tools such as cameras, CCTV, smartphones, and scanners. These images are stored in a database and used for various purposes in the following domains: scientific publications, invoices, multimedia security, forensics, and document verification. The advancement of image manipulation techniques has facilitated the delivery of forged information by allowing the tampering of image contents without leaving obvious traces in the images. There are several opportunities to compromise the last remnants of evidence when using

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different image-altering programs like Photoshop, Paint, and Picasa. Due to the laborious nature of the human process, automatic computer-assisted technologies are essential for image verification [1, 2] and fraudulent document detection to determine the authenticity of a document or identity card [3].

Among the various techniques [5] employed in manipulating images, the term ‘copy-move’ denotes the generation of fresh content within an image by extracting a segment from the same image. Essentially, this involves replicating a specific region within the image. Copy-move is typically utilized to augment the count of pre-existing objects by obscuring them with a portion of the image background. As the characteristics in the image, including illumination, proportion, and focus remain unaffected, such images are more likely to exhibit no discernible evidence of tampering [6]. Two frequently employed methods in the detection of ‘copy-move’ forgery are: the block-based method and the keypoint-based method [8, 9] are commonly employed. The literature presents diverse techniques for identifying forgery in an image, such as block-based DCT for feature extraction [10] and local binary pattern for detection [11]. To facilitate clustering and matching, several algorithms are employed, like K-means clustering and FANN [12, 13]. These approaches aim to identify forged content without relying on a reference image for detection, hence earning the designation of single-image forgery detection methods. Instead of a block-based approach for feature extraction that increases the computation cost [24], a single-image forgery detection method is proposed. Keypoints are extracted using the AKAZE. Next, features are matched using a brute force algorithm. Then, to correctly identify cloned regions BIRCH clustering is applied.

The paper’s organization is as follows, Sect. 2 focuses on related work, Sect. 3 deals with the proposed methodology, and Sect. 4 describes results and discussion. At last, the paper concludes in Sect. 5.

2 Related Work

Parveen et al. [10] introduced a block-based approach for detecting copy-move forgery utilizing discrete cosine transformation (DCT). In this method, a gray image is partitioned into overlapping 8×8 blocks, and features are extracted using DCT across different feature sets. Subsequently, the K-means clustering algorithm is applied to group these blocks. While radix sort is employed to match features, the use of the clustering method enhances the speed of matching in block matching, albeit at the cost of increased time for detecting forged parts in an image. However, the model encounters limitations when faced with images containing multiple forged regions. Al-Qershi et al. [21] presented an approach that involves the use of overlapping blocks, and features are clustered through the K-means algorithm. The RANSAC method is applied to eliminate outliers, and subsequently, the results are identified using a binary detection map. Next, Alberry et al. [22] presented manipulation detection method using Fast SIFT techniques for forensic images from the MICC-220 dataset. Here, Fuzzy C Means clustering is used to detect the forgery portion. Next, the forgery

detection method proposed by Soni et al. [17] studies various block-based techniques such as singular value decomposition (SVD), PCA, FFT, and experiments carried out on various datasets with scaling and rotational attacks. Priyanka et al. [20] presented a novice technique for identifying copy-move forgery. The method combines DCT and SVD to enhance robustness against compression, transformations, and noise. An SVM classifies images as authentic or forged, while K-means clustering localizes forged regions. The method surpasses other state-of-the-art techniques in accuracy, precision, recall, and F1-score for forgery detection. Jaiswal et al. [25] introduced a technique employing shift invariant stationary wavelet transform (SWT) and block division mean feature vector in the YCbCr color space. This approach utilizes overlapping blocks for feature extraction, with the blocks further subdivided into four rectangular and two triangular blocks. While the method successfully identifies forgery, it is associated with high computational time. Afterward, Meena et al. [27] introduced a hybrid approach that combines Fourier Mellin for block-based computation with SIFT for a keypoint-based approach. This hybrid method is preferred over using solely SIFT, as the latter struggles to extract keypoints from smooth regions. The image is partitioned into smooth and textured parts, with SIFT applied in the textured region and FMT utilized in the smooth region. The matching of keypoints and blocks is achieved through G2NN and PatchMatch algorithms, respectively. Furthermore keypoints-based approaches are followed like, Wang et al. [16] introduced a novel approach that combines accelerated-KAZE (AKAZE) and speeded-up robust features (SURF) for forgery detection. Many keypoint-based forgery detection methods face challenges in acquiring sufficient points in smoother regions. To overcome this limitation, the proposed method establishes low response thresholds for both the AKAZE and SURF feature detection steps. The innovation extends to the introduction of a correlation coefficient map, which delineates duplicated regions through a fusion of filtering and mathematical morphology operations. Rigorous experiments validate the effectiveness of this method in identifying duplicated areas and its resilience against various distortions and post-processing techniques—such as noise injection, rotation, scaling, image blurring, JPEG compression, and hybrid image manipulation. Significantly, the experimental results highlight the superiority of the proposed approach compared with other tested copy-move forgery detection methods.

The methods mentioned above perform effectively in restricted constraints such as duplication of the object limited to one or two, dynamic range of intensity values in the image is limited, presence of outliers leads to wrong results, and in certain cases computational time is high.

3 Methodology

The proposed approach uses a fusion of methods to identify the forged region inside an image without utilizing a reference image, the steps are, 5 feature extraction, feature matching, and clustering (Fig. 1).

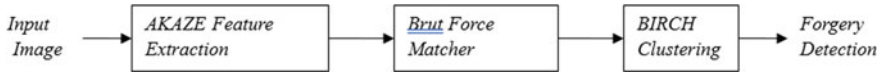


Fig. 1 Diagram outlining the proposed method

Minimal computations are required for recognizing objects in grayscale images compared with color images. It has been observed that grayscale images yield more accurate results than RGB images [14, 15]. Therefore, an RGB image is transfigured to a gray scale using the frequent method by taking a weighted average of color channels which preserves luminance information. The weights are often chosen to mirror how people perceive color intensity, with green being perceived as being the strongest, followed by red and blue. The formula for the weighted average is:

$$\text{GrayValue} = 0.2989 * r + 0.5870 * g + 0.1140 * b \quad (1)$$

where b , g , and r are the respective blue, green, and red color channel values of the pixel, each ranging in the scale of 0–255.

3.1 Feature Extraction

The AKAZE algorithm is used for feature extraction as a choice because of its nonlinear diffusion filtering [32]. However, AKAZE forms its nonlinear scale spaces through an effective framework called fast explicit diffusion (FED) which is comparatively better than scale invariant feature transform (SIFT). AKAZE generates binary descriptors and reduces the higher memory consumption rate [34], whereas SIFT generates floating point descriptors. Even in binarized-SIFT we get binary descriptors but there are chances of losing information from an image.

The AKAZE detector relies on the determinant of the Hessian Matrix and applies Scharr filters to enhance rotation invariance. Feature points are identified by locating maxima in the detector responses at specific spatial locations. The descriptor of AKAZE is generated efficiently using the modified local difference binary (MLDB) algorithm [33]. These AKAZE features exhibit scale and rotation invariance, providing enhanced distinctiveness across various scales due to the application of nonlinear scale spaces [16, 35]. When the AKAZE algorithm is employed on images, it extracts features, and the comparison of descriptor vectors from different regions of the given image enables the identification of similar descriptor vectors. This aids in recognizing and matching the features.

3.2 Feature Matching

The keypoints and descriptors extracted from an image are further taken for matching. The brute force matcher algorithm is applied to features extracted from an image. The method finds correspondences between features in images. For each feature in the image, the algorithm compares its descriptor with all the other features in the image. The goal is to find the best matches based on a similarity metric such as Euclidean distance, Hamming distance, etc., between the feature descriptors based on the keypoints [28]. For the proposed method euclidean distance is considered. Afterward, for each keypoint ' k_i ' in the image, ' D_i ' is the descriptor associated with ' k_i '. Compare ' D_i ' with the descriptors of all other keypoints in the same image and calculate the similarity metric $S(D_i, D_j)$ for each comparison, where ' D_j ' is the descriptor of another keypoint.

To filter out potential false matches, a threshold on the similarity metric is applied, like apply a threshold T to the similarity scores, $S(D_i, D_j) < T$. Here, only matches with $S(D_i, D_j) < T$ are considered valid correspondences. The output of this process is a list of matched keypoint pairs $\{(k_1, k_2), (k_3, k_4), \dots, (k_{n-1}, k_n)\}$, where each pair contains two keypoints from the same image that are considered to be corresponding based on their similarity in terms of their descriptors.

3.3 Clustering

From the previous step, the feature matching process extracted matching data points that are subject to further analysis.

After identifying the matching pairs, a clustering algorithm is applied to the spatial positions of keypoints. This algorithm groups closely located keypoints, revealing clustered sections that helps to identify the forged area within an image. The features undergo clustering using the BIRCH clustering algorithm. BIRCH produces a hierarchical clustering structure, which allows exploring clusters at different granularity levels. It can handle clusters of varying shapes and densities. It constructs a hierarchical cluster tree called the CF-tree (clustering feature tree) to represent the data in a hierarchical manner. BIRCH uses a density-based clustering method internally to create and update this tree structure. The hierarchical aspect of BIRCH comes from the way it organizes subclusters into a tree, allowing for hierarchical exploration of the data. However, the density-based nature of BIRCH is used within the construction and updating process of the CF-tree to merge subclusters based on a distance criterion. This density-based merging helps BIRCH to handle clusters of varying shapes and densities efficiently. BIRCH is generally faster than traditional clustering algorithms like K-means, especially when dealing with large datasets. Due to its ability to incrementally update clusters and its memory efficiency.

The Birch clustering algorithm is used to cluster two sets of points. It combines 'points_1' and 'points_2' into a single array called 'points'. And then creates a birch

clustering model with the given ‘threshold’ and ‘branching_factor’ [23]. Next, fit the birch model to the data (points) to get cluster labels. Then filter out noise points (cluster label-1) and get valid points and labels. Divide the valid points array into two sets (‘valid_points[:n]’ and ‘valid_points[n:]’), where ‘n’ is half of the total number of points. This step provides the required output as detecting forgery from the tampered image. If no match is found between matching features then that image is returned as the original image.

4 Results and Discussions

The experiments were conducted on publicly available datasets, viz., MICC-F220, MICC-F600, MICC-F2000, and CoMoFoD. The MICC-F220 dataset is a well-balanced collection of 220 images, including 110 original and 110 forged images. These images, sourced from the ‘Columbia photographic image repository [2]’, have resolutions ranging from 722×480 to 800×600 pixels. The tampered area constitutes 1.2% of the image, with copied portions randomly placed in either rectangular or square shapes. Forged attacks on the images involve rotations or scaling. The MICC-F600 dataset comprises 440 original images and 160 tampered images. Image resolutions vary from 800×532 to 3888×2592 pixels. The size of the tampered regions differs across images [26]. This dataset includes binary masks of the forged images for result analysis, even it indicates multiple copied regions within the same image. The MICC-F2000 dataset comprises 1300 genuine images and 700 forged images, all with a resolution of 739×492 [13]. This dataset is employed to assess the resilience of the proposed method against geometric attacks, encompassing translation, rotation, and stretching, as well as various combinations of these operations. The algorithm’s robustness is evaluated based on the degrees of rotation, stretch, and translation, each imposing distinct requirements on its performance. CoMoFoD dataset contains a total 10,400 images of size 512×512 which includes post-processing transformation on tampered images like JPEG compression, image blurring, color reduction, and contrast adjustment. Along with that color mask and binary mask are given for ground truth comparison [31]. The experiments are performed on various datasets with various approaches. And Fig. 2 represents results of the proposed methodology on MICC-F220.

Density-based and hierarchical-based clustering method generally includes false matching points in some of the images. A comparative chart with different approaches from the literature in terms of the accuracy of the methods is presented in Table 1.

From Table 1 it is evident that various methods yielded lower accuracy due to the existence of outliers in the image and false positive results. However, the method proposed using AKAZE and BIRCH yields good accuracy. Experiments are carried out on images from MICC-F600, MICC-F2000 datasets, and CoMoFoD. The extent of post-processing attacks in the MICC-F220 dataset is less extensive compared with the MICC-F2000 dataset. Specifically, the MICC-F200 dataset features only a limited

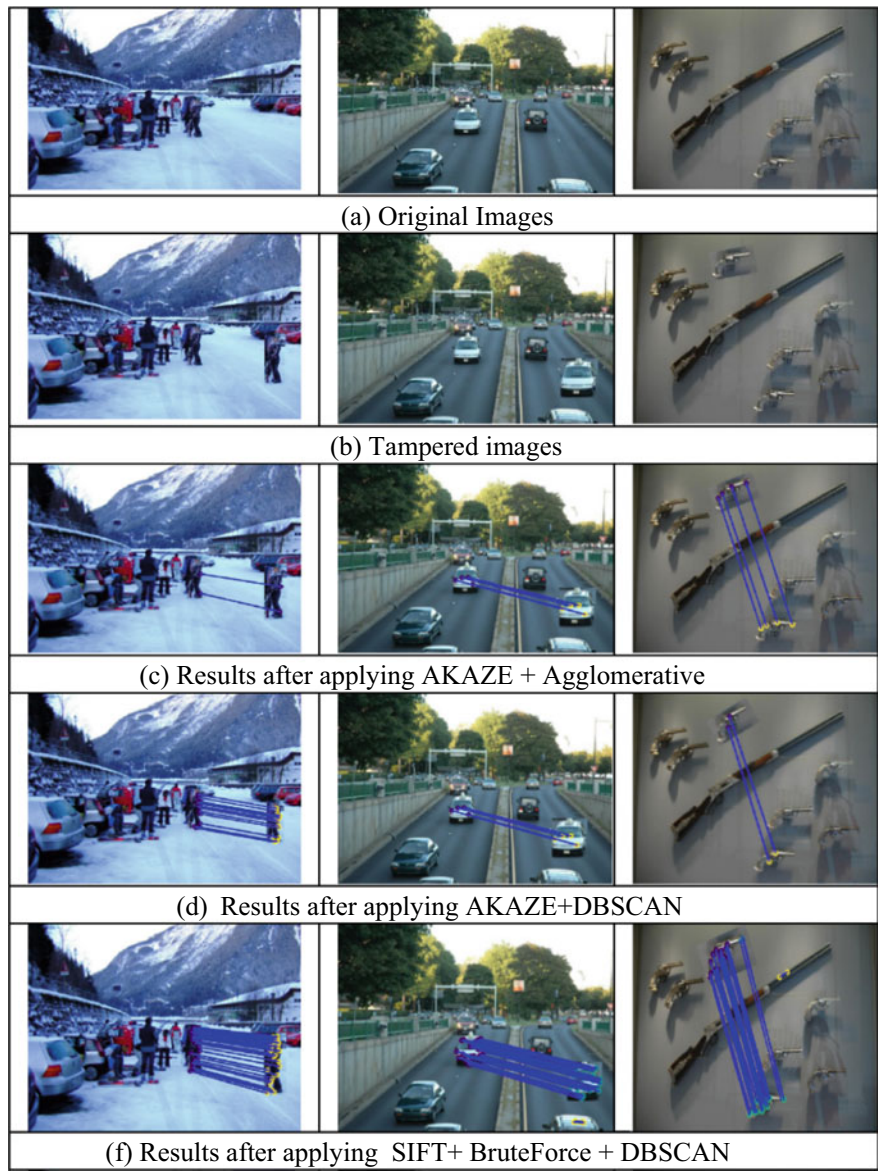


Fig. 2 Sample images from MICC-F220 dataset

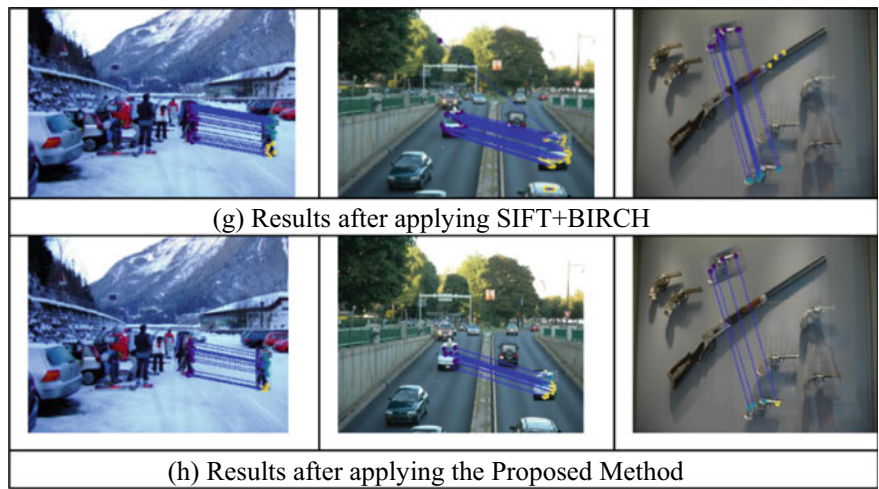


Fig. 2 (continued)

Table 1 Comparison for the MICC-F220 dataset

Author	Methodology	Accuracy in %
Khan et al. (2018) [11]	Hybrid with MinEigen	90
Elaskily et al. (2020) [26]	CNN (15 epochs)	92.18
Vaishnavi et al. (2019) [29]	Symmetry features	83.64
Gulivindala et al. (2016) [37]	DWT + SIFT + PCA	64
Navdeep et al. (2019) [4]	FFT + DCT + SVM	88.62
Gulnawaz et al. (2020) [30]	DCT + kdtree	79.33
Guzin et al. (2016) [8]	KAZE feature	80
Fan et al. (2017) [34]	KAZE and SIFT	78.33
Diaa et al. (2015) [36]	Hessian features	92
Rajput et al. (2023) [19]	SIFT + DBSCAN	92
Proposed method	AKAZE + BIRCH	96

amount of equal stretching, whereas MICC-F2000 incorporates functions involving unequal stretching and combinations [13]. CoMoFoD dataset has 200 image sets along with 40 images per transformation type [31]. The proposed method detects an accurate forged region (Fig. 3).

Figure 3a shows the results from the MICC-F600 dataset, which detected small forged regions with the least amount of false positives when compared with the dataset’s ground truth binary mask. Figure 3b shows robust results for MICC-F2000 dataset, some sample images are given in figure and Fig. 3c shows results for the CoMoFoD dataset. The proposed method is giving better results as we can see in the mask image. The mask image contains original and tampered area of the tampered

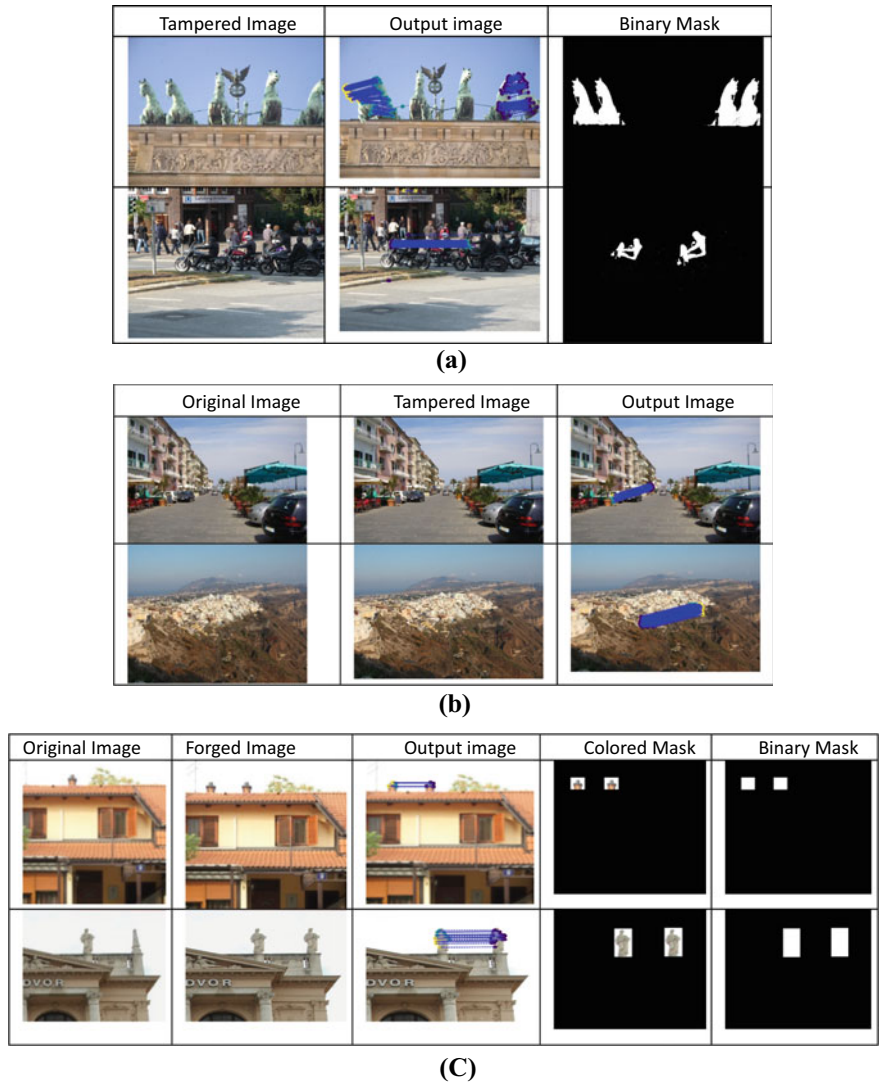


Fig. 3 **a** Results of sample images from MICC-F600 dataset. **b** Results for MICC-F2000 dataset. **c** Results of sample images from CoMoFoD dataset

image. The proposed approach does not use any of the reference images for copy move forgery region detection.

5 Conclusion

Utilizing keypoints is a reliable and efficient strategy for identifying copy-move forgery. The AKAZE algorithm is employed to generate keypoints and descriptors, which are subsequently matched using the brute force matching method. These matched descriptors are then clustered using BIRCH, where the detection of forgery relies on the similarity of legitimate clusters. The suggested approach demonstrates high accuracy on the MICC-F220 dataset. The MICC-F2000 dataset, characterized by extensive post-processing attacks like scaling and rotation, shows improved results. The same method provides higher accuracy for large datasets of CoMoFoD with proper detection of tampered areas. Even the proposed method works fine with MICC-F600 dataset images. MICC-F600 Dataset has sudden pixel variations which causes improper detection for relatively few numbers of images. Therefore, in the future, there is scope to work on images with non-uniform backgrounds with minute changes for forgery detection.

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